

1. Which WAN technologies or protocols (e.g., MPLS, broadband VPN, SD-WAN) could reduce costs compared to leased lines?
 2. What trade-offs in performance and reliability might result from choosing a lower-cost WAN solution?
 3. How could QoS be implemented to ensure acceptable video meeting quality over a cost-optimized WAN?
-

Scenario 15: QoS and Application Prioritization

A healthcare organization connects clinics to a central data center over a WAN. The network carries electronic medical records, VoIP calls, and real-time video consultations. During peak hours, voice quality degrades.

Questions:

1. Which WAN protocols or technologies are best suited for supporting real-time traffic?
 2. How should QoS policies be designed to prioritize voice and video over data traffic?
 3. What risks to reliability and patient care could occur if QoS is misconfigured or unavailable?
-

Scenario 16: Reliability and Redundancy in WAN Design

An international retail company relies on its WAN for real-time inventory updates and payment processing. Any network outage results in significant financial losses.

Questions:

1. What WAN design strategies can be used to improve reliability and minimize downtime?
2. How can redundancy and failover be implemented using different WAN links or providers?
3. How do cost considerations influence decisions about redundancy and high availability in WAN design?

Questions:

1. Why does MPLS use **labels** instead of traditional IP address lookups at every hop?
 2. How does label switching improve forwarding speed and efficiency in the provider's core network?
 3. What could happen to packet latency if MPLS labels were removed and only IP routing were used?
-

Scenario 11: Traffic Engineering & Quality of Service (QoS)

An enterprise uses MPLS to connect branch offices and runs VoIP, video conferencing, and standard data applications over the same links.

Questions:

1. How does MPLS traffic engineering help guarantee performance for VoIP and video traffic?
 2. Why is MPLS more effective than traditional IP routing for enforcing QoS policies?
 3. What might occur if QoS is not properly configured in an MPLS network during peak traffic hours?
-

Scenario 12: MPLS VPNs & Network Scalability

A service provider offers MPLS Layer 3 VPN services to multiple customers, each with overlapping private IP address ranges.

Questions:

1. How does MPLS allow multiple customers to use the same IP address space without conflict?
 2. What role do **labels** and **VRFs (Virtual Routing and Forwarding tables)** play in customer traffic separation?
 3. What impact would a core MPLS router failure have on customer VPN connectivity?
-

Scenario 13: Cost vs Performance WAN Design

A medium-sized company has a headquarters and four branch offices located in different cities. The company currently uses leased lines, which provide high reliability but are expensive. Management wants to reduce WAN costs while still supporting email, file transfers, and occasional video meetings.

Questions:

Scenario 7: Fixed Frame Structure & Synchronous Transmission

A telecommunications provider uses a SONET network to carry voice calls, live video streams, and high-speed data services over long distances.

Questions:

1. Why does SONET use a **fixed, synchronous frame structure** instead of variable-length frames?
 2. How does SONET's synchronous nature help support real-time services like voice and video?
 3. What impact would synchronization problems have on delay and overall network performance?
-

Scenario 8: Bandwidth Allocation & Service Types

A large enterprise connects its regional offices using a SONET backbone. Some links carry constant voice traffic, while others carry bursty data traffic.

Questions:

1. How does SONET support different bandwidth requirements for voice and data traffic?
 2. Why is SONET well-suited for applications that require **guaranteed bandwidth**?
 3. What could happen if multiple applications attempt to exceed their allocated SONET bandwidth?
-

Scenario 9: Protection Switching & Network Reliability

A fiber cut occurs on a SONET ring connecting several cities. Despite the failure, users experience little to no service disruption.

Questions:

1. How does SONET protection switching help maintain communication during a link failure?
 2. What is the role of **ring topology** (such as UPSR or BLSR) in fault tolerance?
 3. Why is fast recovery (typically within 50 ms) critical for real-time applications?
-

Scenario 10: Label Switching & Packet Forwarding

A service provider deploys an MPLS backbone to carry voice, video, and data traffic between multiple customer sites.

Questions:

1. What does the BECN bit indicate to the receiving router?
2. What does the FECN bit indicate to the remote router?
3. What action should the routers take when these bits are detected?

Scenario 4: ATM Cells and Fixed Cell Size

A financial institution uses an ATM network to support real-time trading applications, voice calls, and regular data transfers over the same infrastructure.

Questions:

1. Why does ATM use a fixed cell size of **53 bytes** instead of variable-length frames?
 2. How does the fixed cell size benefit real-time traffic such as voice and video?
 3. What impact would larger cell sizes have on delay and jitter for voice traffic?
-

Scenario 5: Quality of Service (QoS) and Traffic Classes

A university deploys an ATM backbone to handle video lectures, VoIP calls, and standard web traffic. The network administrator configures different ATM service categories for each type of traffic.

Questions:

1. Which ATM service category (CBR, VBR, ABR, UBR) would be most appropriate for video lectures and why?
 2. Why would VoIP traffic require a different service category than standard web traffic?
 3. How does ATM ensure bandwidth and delay guarantees for high-priority applications?
-

Scenario 6: Virtual Paths (VP) and Virtual Channels (VC)

An ISP uses ATM to connect multiple customers over a shared infrastructure. Each customer runs several applications simultaneously, such as voice, video, and data.

Questions:

1. How do **Virtual Paths (VPs)** and **Virtual Channels (VCs)** help manage traffic from multiple customers?
2. Why might an ISP bundle multiple VCs into a single VP?
3. What happens to active VCs if a VP fails?



DEPARTMENT OF ELECTRONIC AND TELECOMMUNICATION ENGINEERING

Assignment 3, in your groups attempt the following question for presentation. Next week 30th Jan 2026 Friday

Scenario 1: Virtual Circuits & DLCIs

A company has three branch offices (A, B, and C) connected through a Frame Relay cloud. Office A communicates with Office B using DLCI 102 and with Office C using DLCI 103. Office B uses DLCI 201 to communicate with Office A.

Questions:

1. Why do Office A and Office B use different DLCI numbers for the same virtual circuit?
2. What type of virtual circuit is most likely being used in this scenario?
3. If Office C is disconnected, will communication between Office A and B be affected? Explain why or why not.

Scenario 2: CIR, Congestion & DE Bit

An organization subscribes to a Frame Relay service with a **CIR of 256 kbps**. During peak hours, users complain of slow network performance. Network monitoring shows that traffic often exceeds the CIR.

Questions:

1. What happens to frames sent above the CIR during network congestion?
2. How does the **Discard Eligibility (DE) bit** affect these frames?
3. Why might performance appear normal during off-peak hours but degrade during peak hours?

Scenario 3: FECN & BECN Congestion Control

A router connected to a Frame Relay network receives frames with the **BECN bit set**. Shortly after, the remote router reports receiving frames with the **FECN bit set**.